

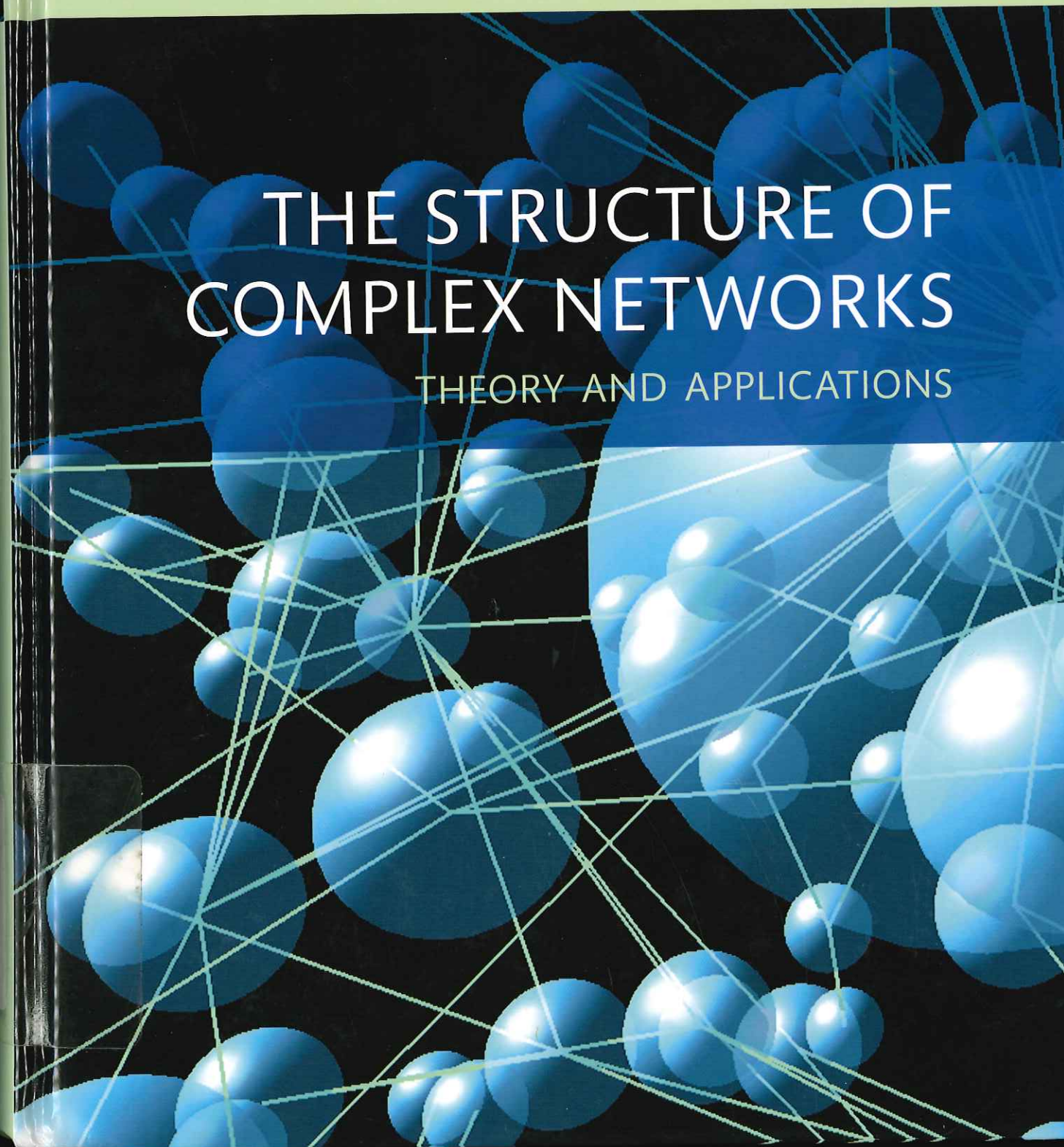
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ERNESTO ESTRADA

THE STRUCTURE OF COMPLEX NETWORKS

THEORY AND APPLICATIONS





This book deals with the analysis of the structure of complex networks by combining results from graph theory, physics, and pattern recognition. The book is divided into two parts. 12 chapters are dedicated to the development of theoretical tools for the structural analysis of networks, and 7 chapters illustrate, in a critical way, applications of these tools to real-world scenarios. The first chapters provide detailed coverage of adjacency as well as metric and topological properties of networks, followed by chapters devoted to the analysis of individual fragments and fragment-based global invariants in complex networks. Analysis of the concepts of communicability, centrality, bipartivity, expansibility, and communities in networks follow. The second part of this book is devoted to the analysis of genetic, protein residue, protein-protein interaction, intercellular, ecological, and socio-economic networks, including important breakthroughs as well as examples of the misuse of structural concepts.

Ernesto Estrada is Full Professor and Chair in Complexity Science, Department of Mathematics and Department of Physics, University of Strathclyde, Glasgow.

'The *Structure of Complex Networks* by E. Estrada is a great book, written by an expert in this new and exciting field. Starting from the basic concepts, it describes the most recent advancements in the field in an open and enjoyable way. I recommend this book to both beginners and practitioners as a valuable tool for their research.'

Guido Caldarelli, ISC-CNR Rome, Italy

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